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AI Based Solution for Autonomous Well Control Leveraging Continuous Multiphase Flow Measurement at Wellhead - Innovative Practice to Knob Remote Operations Challenges

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Abstract

This paper details an innovative practice and adapted initiative implemented by Company Asset to ensure continuous measurement, monitoring, and control of oil flow from oil-producing well leveraging Artificial Intelligence. In addition to maintaining the specified oil flow target, autonomous gas flow control was also effectively managed to ensure comprehensive well control. This arrangement proved advantageous in achieving the well testing key performance indicators (KPIs) and eliminating the need for manual intervention, thereby saving man-hours.

The need for real-time flow monitoring & control at individual wellheads led to evaluating various solutions. A Multiphase Flow Meter (MPFM) was installed, enhancing flow measurement and allowing configuration of oil flow rate signals to control valve. This setup enabled autonomous control of the well, managing both natural oil flow and artificial lift gas flow when needed. The design allows easy relocation based on well-activated needs.

Steps taken included:

- Implementing management of change for new solutions

- Conducting risk assessments

- Evaluating vital GL well surface and subsurface parameters

- Developing safe operating procedures, scenarios, schematics, and a user-friendly control scheme

The autonomous control of the well enabled comprehensive management of the lift gas well. It also significantly reduced the manual labor required for diverting the well into testing facilities. Additionally, it facilitated the optimization of the gas flow rate and ensured the continuous production of oil through the oil-producing wells.

The observed well control behavior was smooth and adhered to the stipulated setpoints. By leveraging the capability to connect the MPFM with the piping connections available at the wellhead, the following benefits were achieved and will continue to be advantageous in the future:

- Optimization of manhours approximately 87 hours for 10 wells / 1,200 hours for 140 wells

- Production gain of 1,850,000 barrels of oil by preventing production loss from gas lift wells
- Optimization of GL flow rate, leading to savings of lean gas (9,000 MMscf for 50 wells) valued at 133 MMUSD
- Achievement of 100% well testing KPIs
- Effective and timely dual-string well testing resulting in a 10 MMUSD production gain
- Ensuring compliance with subsurface parameter restrictions, thereby preserving reservoir health
- Enhancement of well digitalization
- Facilitation of autonomous well control
- Real-time well testing
- Accurate representation of the well's behavior
- AI-based decision-making ensuring adherence to restrictions and WH/reservoir guidelines

The implemented solution is highly effective in ensuring continuous production, timely well testing, and reducing manhours by leveraging autonomous control integrated with artificial intelligence. This methodology can be replicated across other existing facilities within the company and globally for the following reasons:

- Real-time flow monitoring and control
- Well testing compliance
- Effective reservoir management

This implementation will enhance well availability with continuous production while ensuring the health of the reservoir and surface facilities.

Introduction

Gas lift is one of several processes used to artificially lift oil or water from wells where there is insufficient reservoir pressure to produce the well.

The lift gas is provided from compressor station. The gas lift let-down station is supplied with the gas from station where the pressure is let down from 5,800 to 2,610 psig and fed into a dedicated buried network to the gas lift oil producer wells completed in various part of the field.

Gas lift main header and the sub headers supplying various gas lift wells have also provisions to hook up temporary pigging facilities to carry out piping internal inspection.

The gas is injected into the well at a controlled pressure. A motor operated choke valve controls the pressure of the gas at the well. Inlet the wellhead annulus pressure (pressure downstream of the gas lift choke valve) varies from 1000 psig to 2200 psig max.

Field Background and Challenges

The gathering and production facilities include Remote Degassing Stations (RDS) and Central Degassing Station (CDS). The purpose of surface facilities at remote stations is to gather the crude from the wells and transport the crude to the nearest existing processing facility through transfer line. Multiphase Pumps (MPP) are installed at Gathering station to transport multiphase fluid through separate transfer line. The proposed scheme for remote field consists of Oil Gathering System to gather oil from multiple number of production wells to Multiphase Pumping (MPP) station through multi-port selector manifold (MSM). Multi-

Phase Flow Meter (MPFM) will be provided for the well testing. The Multiphase Pump boosts the pressure of the production from wells to a level sufficient to deliver to the transfer line, approximately 21 Km away.

The well fluids from the well are gathered and collected in three remote degassing station RDSs. From the RDS the fluids are transferred through pig gable transfer lines to Central degassing stations for further processing.

The remote stations are manned during the daylight hours and are unmanned during night hours. The remote stations are operated using a DCS and can be operated from CDS control room after daylight hours using the "Remote" option using SCADA.

Wells are categorized as single completion, dual completion or artificial lift.

The reservoir sustainable production rate is maintained by using water injection (water injector wells) and gas injection (gas injector wells).

All wells are operated in accordance with the Well Integrity and Operations Standard. All well operations are planned and performed in compliance with all the applicable legislation and regulations. These may include but are not limited to directives/policy on HSE Performances, Ethical Conduct, Employees, Control and Finance.

Wells are further classified as oil producers, water supply, water injection, gas injection, aquifer reservoir observation, and disposal.

Wells Structure

Wells are equipped with surface safety valve (SSV) and Subsurface safety valve (SSSV). The Xmas tree is part of the wellhead and comprises a series of valves to allow the outflow of well fluids, and any auxiliary fluids to be controlled. Each well flow rate is regulated by a choke valve. Most choke valves are manually adjustable while other valves are automated for flow changes to be performed from the Local Control Panel / Control Room.

Each well is equipped with a wellhead at the surface level. The wellhead consists of three main parts; Casing Head Housing (CHH), Tubing Head Spool (THS) and Xmas Tree. The CHH and THS each contain valves that allow access to the annulus. The Xmas Tree comprises a series of valves to allow the outflow of well fluids, and any auxiliary fluids to be controlled. The well flow rate is regulated by a choke valve.

Choke valves

Choke valves are either positive or adjustable / motorized. The adjustable chokes can be either manual or automatic which allows the choke size change to be performed from the local control panel, RTU or the control room.



Figure 1—Typical Positive Choke Valve



Figure 2—Typical Adjustable Choke Valve



Figure 3—Motorized Choke Valve for oil and gas sides

Positive type chokes incorporate a fixed bean (similar to an orifice plate) that is available in a range of sizes graduated in sixty-fourths of an inch.

If there is a requirement to change the flow rate of a particular well it is necessary to remove the existing bean and replace it with one of a different size. Manual intervention is required.

As the bean in a choke can be abraded by solids in the well fluids (which would increase the size of the bean) there is a requirement to monitor bean performance as related to well flowrate and replace the bean from time to time.

Adjustable choke valves are installed in order to provide the means of flow regulation from the wells.

Most wells are completed with four casing strings.

The surface casing of 18 5/8 inch or 20 inch diameter, called conductor, is usually set to a below surface depth lower than 300 ft.

Set inside the surface casing is the 13 3/8 inch diameter casing to an approximate depth of 5000 ft, and a further 9 5/8 inch casing to a depth of about 8000 ft.

A 7 inch or 4 1/2 inch diameter liner hangs inside the 9 5/8 inch casing 100 ft from the bottom and set across the pay zone.

The space between two sets of casing is called annulus.

The purpose of the surface and intermediate casing strings is to seal the well, preventing any oil or gas leakage through the ground to the surface or into the aquifers.

Annulus A-B-C are present.

Annulus A: the annulus between the tubing and production casing.

Annulus B: the annulus between the production and the intermediate casing.

Annulus C: the annulus between the intermediate casing and surface casing.

Wells are categorized as:

Single completion (single tubing string);

Dual completion (two tubing strings);

Artificial lift using electrical submersible pumps or gas lift.

Surface arrangements for each well are generally similar.

Artificial lift Wells

Artificial Lift can be carried out using Electric Submersible Pump (ESP) or lift gas.

The Artificial Lift producers using ESP are single string and equipped with a down hole Electric Submersible Pump (ESP) to boost the flow from the original free flowing wells.

The principle of gas lift is that gas injected into the tubing reduces the density of the fluids in the tubing, and the bubbles have a "scrubbing" action on the liquids. The single completion gas lift as single completion well have one tubing for the oil extraction from the reservoir for the oil sending to the gathering manifold and one string for gas injection.

Wellheads

The wellhead is located on the top of the well casing strings and provides a seal for the annuli created by the different sized casings within the well, all of which terminate at approximately ground level.

On the casing head housing and tubing head spool of the wellhead various valves are located.

Dedicate valves provide access to the annuli created by the casing strings. Access is required to test the pressure build up in the annuli, to take fluid samples from the annuli and, when necessary, to bleed off pressure. Valves are often installed in tandem to provide positive shut-off. The termination point is either blanked, or may be fitted with a needle valve and plugged off.

The Xmas tree incorporates five main valves and a wellhead cap for each string of the well. The valves are always gate-type valves.

The lowest valve is called the Lower Master Valve which, to avoid erosion, is only used in emergency situation, serves to isolate well fluids from the Xmas tree and must never be used for routine isolation of a well.

Above the lower master valve is the Upper Master Valve.

On the production tubing, above the Upper Master Valve, one or two Wing Valves are provided. One is the start of the well flow line.

On the top of the production tubing is the Swab Valve. This is used for wireline operations to provide isolation between the wireline equipment (the lubricator) and the well. The Bottom Hole Test Adaptor (BHTA) cap is removed (with the swab valve closed) and the wireline lubricator is attached in place of the cap. The lubricator provides a means of ingress and egress to the well tubing strings while the well remains under pressure.

Wellhead Control Panel

All producer wells have wellhead control panels. The SC-SSSV and SC-SSV close due to:

- Low pressure on actuation of low pressure pilot;
- High pressure on actuation of high pressure pilot;
- Detection of fire near the wellhead by melting the fire fusible plugs: one high pressure fire fusible plug installed in the hydraulic line to the SC-SSSV and four low pressure fusible plugs.

The instrumentation associated with the wellheads and flow lines includes local pressure and temperature gauges.

General Operating philosophy and requirements

Opening of Valves

Opening of the valves is a manual operation. Under no circumstances any of the valves open automatically. The operating sequence is designed such that SC-SSSV always open before SSV(s). The SSV(s) can be manually opened and closed (locally and remotely) as long as the SC-SSSV is opened. All the actuators are spring loaded to close type (FTC; fail to close).

Control system design logic ensures that the above opening sequence is always maintained. Change in the opening sequence is not permitted.

The panel incorporates start-up bypass requirement to bypass low pressure signal. The bypass system is auto reset when the low pressure signal has been cleared. High/Low pressure status indicators are provided at the panel front.

The Wellhead Control Panel is used to control Surface Safety Valve(s) and Surface Controlled Sub-surface Safety Valve by applying pressurized hydraulic oil (pressure to open) to the valves' actuators. The SSV(s) and SSSV are fail closed valves and are normally kept in the "Open" position by energized low powered directional solenoids. De-energizing the solenoids will redirect the pressurized oil from the valves' actuators. The oil pressure shall be generated by an electric motor driven pump. Motor power rating shall be sufficient to generate the required pressure for opening the above valves.

The wellhead control system is totally independent of the flow line fluid using clean hydraulic oil in a closed, leak tight circuit. This is required to eliminate the risk of contamination of the actuator and control circuit by well fluid and to ensure reliability and durability of WHCP function.

The SSV(s) & the SC-SSSV usually operate at different pressures. Two pressure ranges are provided for hydraulic supply, one pressure range for SSV(s) & second pressure range for SC-SSSV.

The wellhead control system allows manual operation of SC-SSSV and SSV(s). The controls are designed for fail safe operations and easy to service without removing the hydraulic relays out of the system for maintenance and calibration. System is designed both for user and environment friendliness.

The control panel is solar powered designed as a stand-alone wellhead control system both for the operation of single or dual completion oil producers as given in the material requisition.

Control system design logic ensures that the above opening sequence is always maintained.

The SSSV mounted on the tubing of the wellhead, is hydraulically actuated and is open during normal operation.

A high pressure fusible plug mounted on the hydraulic actuation line allows, when it melts (94°C) in case of fire, the SSSV closing, while around the wellhead four low pressure fusible plugs are mounted by separate headers.

SSSV closes after a time delay and only when the ESD is activated (the ESD push button mounted in front of the panel, remote ESD actuation or fusible plugs actuation).

A remote ESD point is installed at least 30 metres downwind from the wellhead at a suitable location & at well fence gate.

Solar System/Passive Shelter

The power supply for the panel is 24 VDC, -10%, +15%, negative earthed. This power supply is fed by the SOLAR power system supplied by others. The panel receives power by the non-redundant separate feeder for hydraulic pumps and control panel.

The solar system is provided at the wellhead station to meet the power requirement for the operation of the wellhead control panel, PLC and SCADA panels and limited lighting. An isolated, thermally insulated enclosure is provided as a passive shelter for housing the PLC, SCADA panels. The solar system and the passive shelter with the thermo-siphon based water cooling arrangement, which maintains the temperature inside the shelter at the required operating level, day and night, leave the wellhead operation independent of external utilities.

High and low fusible plugs are hydraulically powered by the wellhead panel hydraulic system. Whenever a fire occurs around the well, the hydraulic pressure loop of the LP fusible plugs or the hydraulic pressure loop of the HP fusible plug gets depressurised causing the shutdown of the wellhead.

In addition to the above safety protections, the wellhead is also provided with an Emergency Shutdown switch. A remote ESD point is installed at least at 30 metres downwind from the wellhead at a suitable location & at well fence gate.

Single Completion Gas Lift Well

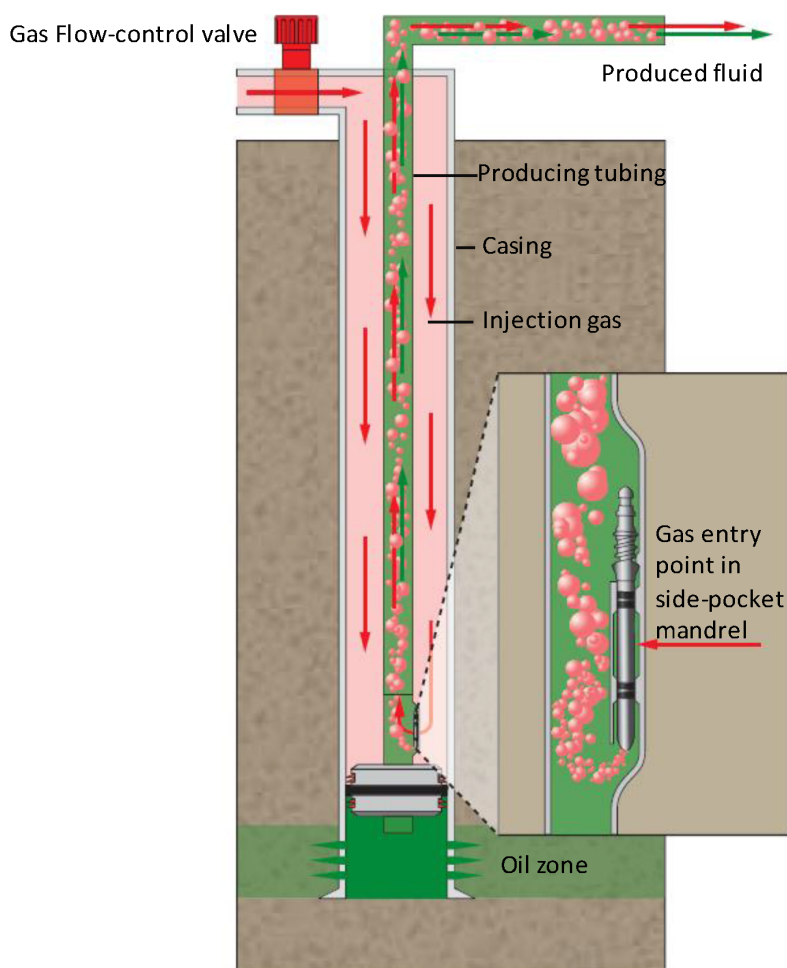


Figure 4—Typical Single Oil production Gas lift Completion Type

Introduction

The principle of gas lift is that gas injected into the tubing reduces the density of the fluids in the tubing, and the bubbles have a "scrubbing" action on the liquids. The single completion gas lift as single completion well have one tubing for the oil extraction from the reservoir for the oil sending to the gathering manifold and one string for gas injection. Unlike the single completion, in the single completion gas lift, oil extraction is carried out by injecting compressed lean gas by the means of a second line (gas lift piping).

Lean gas comes nearby plant to ensure uninterrupted supply of lift gas for the wells.

The lean gas from the compressor discharges is routed to the pressure let-down station for gas lift where pressure is reduced from 400 barg to 180 barg.

The gas is then routed to the wells through a pipeline network including a 6" main header and 4" sub-header with the 3" tap-offs to the individual wells.

Surface arrangement

Gas Side

The wellhead 2" piping with double block valves is located at about 100m away from the wellhead, inside a fixed fenced area, for the well isolation during maintenance activities. Also a shutdown valve (SDV) is provided in order to isolate the pipeline from the wellhead in the event of fire due to the oil spill in the well area and ruptures in the piping inside the wellhead fenced. A Y strainer, a vortex flow meter and a pressure control valve are also provided on the gas side.

The gas flow to the well is regulated at the pressure required for the gas lift. This is achieved by pressure controller and a motorized choke valve.

Double block valves are provided for isolation at the wellheads at the Xmas tree end and at the gas lift piping battery limit end.

A check valve is provided between the manual isolation valve and the gas side wing valve.

The surface safety valve hydraulically actuated, located between the check valve and the wing valve (side gas) is handled from the wellhead control panel.

2" manual depressurising provision for the wellhead gas lift piping is given at the downstream of the SDV, downstream of the remote operated choke valve and downstream of the check valve discharging to the vent boom.

The vent boom is provided with a flame arrestor. It should be noted that this vent is for depressurising the well head piping alone and not for gas flow pipelines and flow line from the gas injection compressor at NGIP, which if required needs to be done inside NGIP limits.

The provision for nitrogen purging of the injection line is also given.

A DPI indicator measures the pressure drop across the Y strainer. The flow meter indication is pressure and temperature compensated. The remote operated choke opening is adjusted based on this flow indication.

Oil Side

The 3" string producer is equipped with wing valve, surface safety valve (SSV) (hydraulically actuated), choke valve for the flow regulation, pressure transmitter and related local pressure indicator (PI), pressure transmitters (PT), check valve (wafer type) to prevent oil back flow from the gathering manifold in case of the well shutdown. The sub surface safety valve (SSSV) installed on the tubing is open in normal operation. Opening/closing (this valve is hydraulically actuated) is handled from the wellhead control panel LCP.

Wellhead control panel LCP

The PLC based wellhead control panel is envisaged with solar power. The wellhead control panel functions are the following:

- To control the opening and closing of the gas lift SDV at the battery limit (SDV), the surface safety valves on gas side (SSV) and oil side (SSV).
- To open subsurface safety valve SSSV.
- To take care of the well safety requirements like fire, gas and ESD activation.
- To provide interface through the data link to the remote NGIP control room for monitoring the well parameters as well as facilitate remote operation of the well.

RTU/PLC XCP

Wellheads have also provision for remote operation of the following actions through the data link between DCS at NGIP and RTU/PLC at the wellheads:

Adjust gas lift choke (gas side choke valve) by the remote set point for gas lift pressure Well ESD command

Well testing facilities

Production from the inlet system at CDS is diverted into five Separator Trains, in parallel configuration. Crude oil from 2nd Stage Production Separators is routed, by a common distribution header to three vertical gas/liquid separators named Degassing Boots.

The production from each local oil well can be routed to one of the new three phases test separators and one MPFM located at CDS plant.

The purpose of a Test Separator is to allow the component flows (oil, gas and water), from a single well or source, to be measured and sampled. Any changes in a well's test results are plotted over a given time to give an indication of the change in reservoir conditions in the area from which the well is flowing.

Test separators are horizontal pressure vessels equipped with an internal weir, which allows settling of water and sludge in the first section of the vessel. Oil overflows the weir and collects in the oil compartment.

Each test separator is provided with two control and metering systems (one for low range and one for high range) in parallel installation for either, gas or liquid outlet flow rates. The operating pressure is the same of the 1st Stage Production Separator (i.e., 17.2 barg).

The operator manually selects from the control room the proper control and metering systems according to the expected ranges of liquid and gas flow from the well under testing.

Testing Facilities

MPFM

The Multiphase Flow Meter (MPFM) is a system to measure a flow rate of multiphase flow (oil, gas and water) and is used for testing well fluids from the Inlet Manifold.

The two installed Multiphase Flow Meters FQI (MPFMs) allow the testing of two wells at a time during normal operation.

Each MPFM is always kept online, even when the testing of wells is not required.

The oil, gas and water flow rates, pressure and temperature of the well tested are transmitted to the BQ Control Room. Local on-site indication is also provided.

After metering the fluid is sent to the inlet of the MPP.

The MPFM is provided to measure the individual oil, water and gas flow rates of three phase crude. The diversion of the well under test to the MPFM is carried out manually. In order to avoid stagnant flow conditions to the header and MPFM, one well is always aligned through the MPFM always.

Temperature and pressure compensation are provided for the gas flow rate. The flow rate and totalized flow indication are available on the control system workstation in CCR.

MPFM draining is performed through the drain line to the drain header.

Double block-and-bleed 6" NPS connections are fitted to allow a portable test separator to be connected in order to verify the MPFM performance.

Meter Arrangement

The Multiphase Flow Metering System is designed to measure the individual flow rates of oil, water and gas (temperature and pressure compensation are provided for the gas flow rate); mass and volume calculations, cumulative measurements and GVF, GLR, GOR and water cut calculations.

The System consists of the following main components:

- Field Sensor Unit;
- Field Control Unit;
- Human-Machine Interface (HMI).
- Calculation is performed in the MPFM HMI; the final data are transferred to the DCS for data logging.
- Double block-and-bleed 6" NPS Portable meter prover connections are available for proving the MPFM if/as required.
- Control Description
- The diversion of the well being tested to the MPFM through one Test Header is performed remotely from Control Room; the operation can be also carried out locally in the field.
- An alarm is generated when the operating point falls beyond the meter envelope.

Field Sensor Unit

The Field Sensor Unit is installed on the MPFM inlet line and measures the parameters used in the flow calculations of oil, gas and water.

The field sensors are suitable for all the following flow regimes:

- Stratified flow;
- Bubbly flow;
- Churn flow;
- Slug flow.

The MPFM is capable of functioning and operating in the stream condition of 0-100% Water in Liquid Ratio (WLR) and 0-99% Gas Void Fraction (GVF) and all flow regimes, unaffected by paraffin or asphaltene deposition.

The in-line MPFM can perform self-verification checks (e.g. continuous system status self-checks, pressure measurement hardware, dynamic self-verification, etc.)

The overall accuracy of MPFM is as follows:

- Gas flow rate $\pm 5\%$ to $\pm 10\%$ relative
- Oil flow rate $\pm 5\%$ to $\pm 10\%$ relative
- Water flow rate $\pm 3\%$ to $\pm 8\%$ absolute

- Total fluid flow rate $\pm 2\%$ of reading
- Repeatability $\pm 5\%$ relative for each of the phases $\pm 5\%$ relative for WLR

Objective

Looking into the following challenges, the thought process resulted into the solution keeping in view the effectiveness of MPFM.

Challenges/ Driver

- Operator intervention for well diversion to testing facility at station
- Commuting to the site.
- No real time flow measurement
- Requirement of gas lift flow optimization and continuous flow
- Lesser frequency of well testing due to limited number of testing facilities

Solution

- Installation of compact MPFM (Flange connections already available at wellhead) – Real time flow monitoring
- Oil flow to work as PV for Oil choke valve controller & GL choke valve

Restrictions

- Ensuring & maintaining 100psi differential across oil choke valve for Natural flow mode
- Ensuring & maintaining 70psi differential across gas choke valve for gas lift mode
- No more lift gas flow than 1.5 MMscfd

Execution Strategy

It was decided to install Multiphase Flow Meter on individual wellheads for the primary purpose of measuring and maintaining the production by controlling the wellhead choke and the Gas lift valve.

The above will be accomplished by tie-in the MPFM to process line and linking the communication to the DCS / SCADA system to utilize the MPFM registers feedback like the oil flow rate and the MPFM differential pressure to send control signals from the DCS / SCADA system to the wellhead choke and the Gas lift valve to increase or decrease opening.

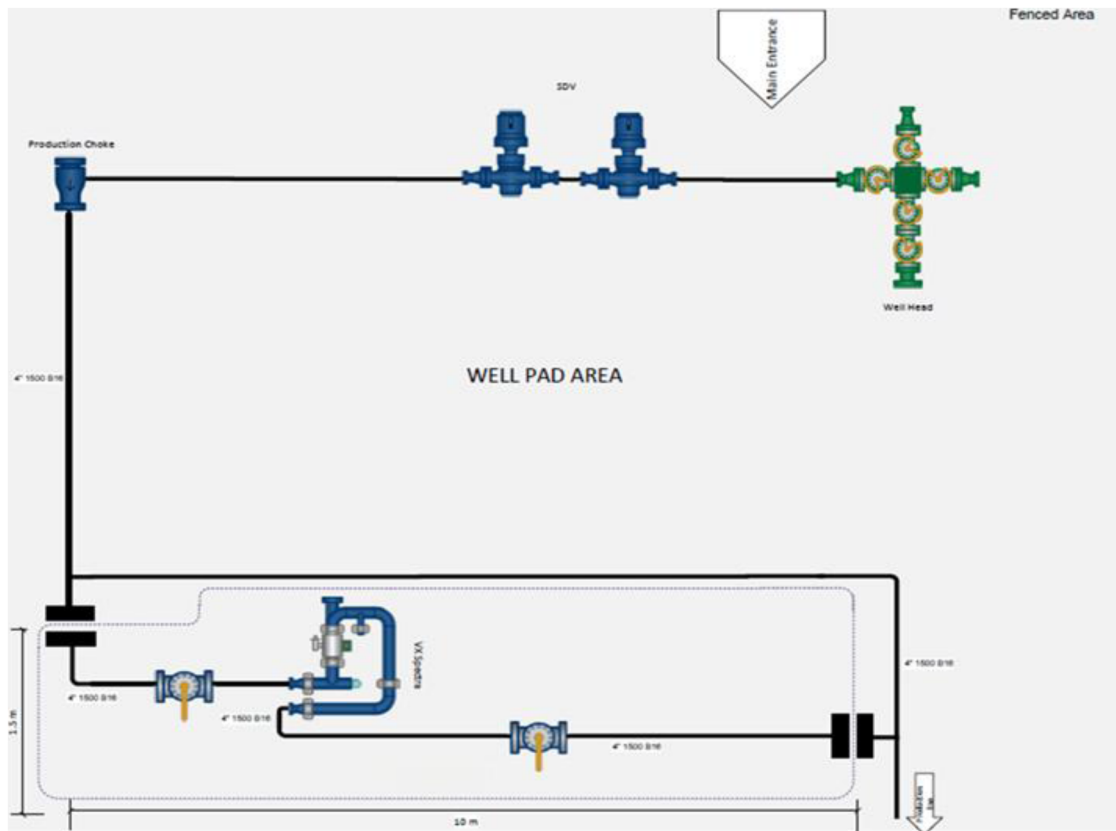


Figure 5—MPFM installation at Wellhead

MPFM Specification

- 4.1 Weight and Footprint
- Weight and Footprint
- 2-inch Single Leg Diligens Spectra
- Length (m) 2
- Height (m) 1.94
- Width (m) 1.05
- Weight (kg) 1350
- Diligens Spectra Operating Ranges
- Working Pressure 1440 psi
- Hydrostatic Test Pressure 2160 psi
- Process Temperature -29° C to 121° C
- Ambient Temperature -20° C to 60° C
- 4.3 Electrical Specifications
- The PSU accepts both AC and DC power sources and converts them to a 24V DC supply for the DAFC of
- Diligence Spectra.

- VX Spectra Operating Range
- DC range 20 - 30V DC
- AC range 100 - 240V AC Diligens Spectra Certifications
- Certifications
- Component Certification
- Diligens Spectra Piping design ASME B31.3

Vx Spectra Certifications

- Certifications
- Component Certification Ex Certification
- Vx Spectra Pressure Equipment Directive (PED),
97/23/EC
- Ex d IIB T4
- 4.6 Hazardous Areas Certification / Approval

The user needs to make sure that Vx Spectra is used below ambient temperature rating of 60° C.
The Vx Spectra has been designed to work in the following Ex Hazardous Areas:

- o Zone 1
- o Zone 2
- o Gas Group IIB
- o Temperature Class T4

MPFM monitoring and Control Logic Block Diagram

MPFM – Logic Block Diagram

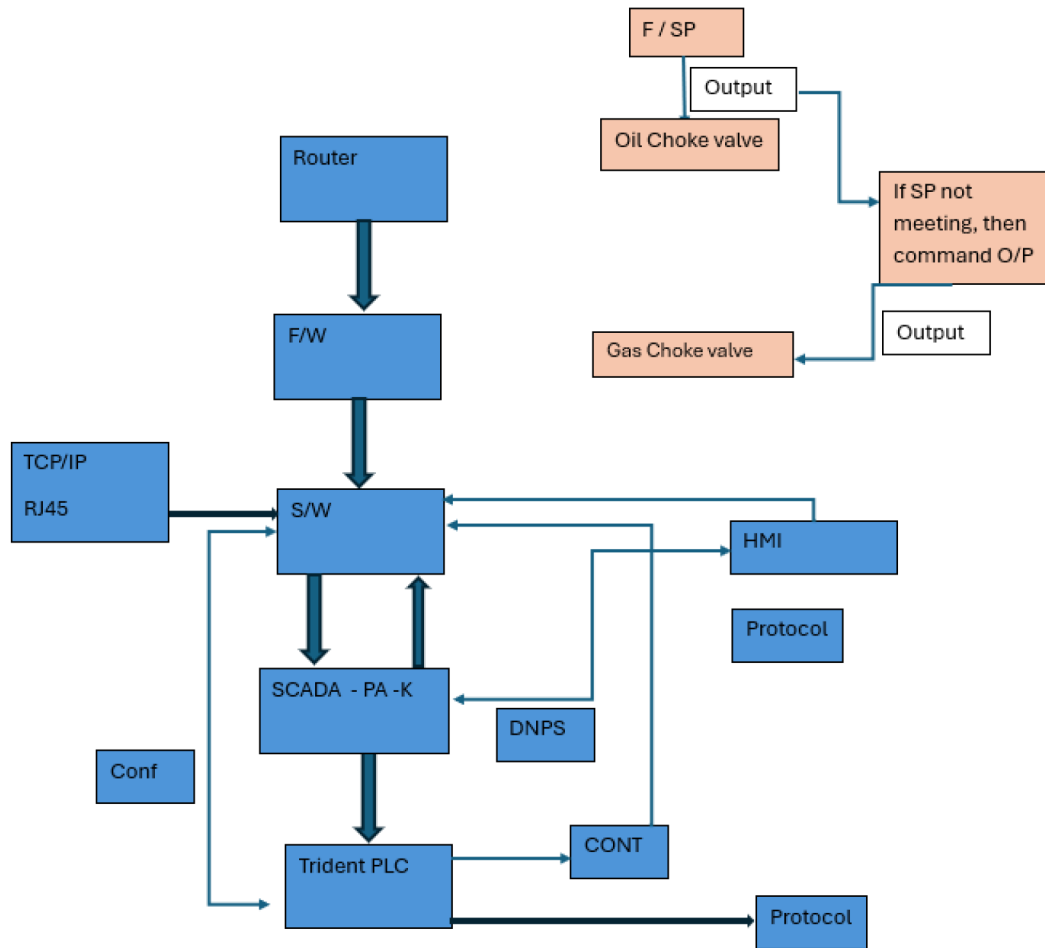


Figure 6—Logic Block Diagram

- Installation of compact MPFM (Flange connections already available at wellhead) – Real time flow measurements and monitoring
- SP to be given for oil flow rate in BOPD
- Real time Oil flow measured from MPFM to work as PV and output % firstly to control Oil choke valve
- If the oil flow SP is not met, the output % to control gas flow through GL choke valve

Restrictions

- Ensuring & maintaining 100psi differential across oil choke valve for Natural flow mode
- Ensuring & maintaining 70psi differential across gas choke valve for gas lift mode
- No more lift gas flow than 1.5 MMscfd
- GOR as per SH guidelines

MPFM Modbus Interface

Serial Link: Modbus RTU

The following lists the requirements for the Modbus RTU serial link interface between the Vx Spectra DAFC (Modbus slave) and the client PCS (Modbus Master).

Serial Port

RS-485 (2-wire)

Baud rate: 1200,2400,4800,9600,19200,38400,57600,115200

Data bits: 8

Parity: Odd, Even or None

Stop bits: 1 or 2.

Flow control: None

Modbus Protocol

Slave Address: 1 to 247

Function codes:

03 – Read Holding Registers

04 – Read Input Registers 05

– Force Single Coil

16 – Preset Multiple Registers

Ethernet: Modbus TCP (Used during field trial)

The following lists the requirement for the Modbus TCP Ethernet interface between the Vx Spectra DAFC (Modbus Slave) and the client PCS (Modbus Master).

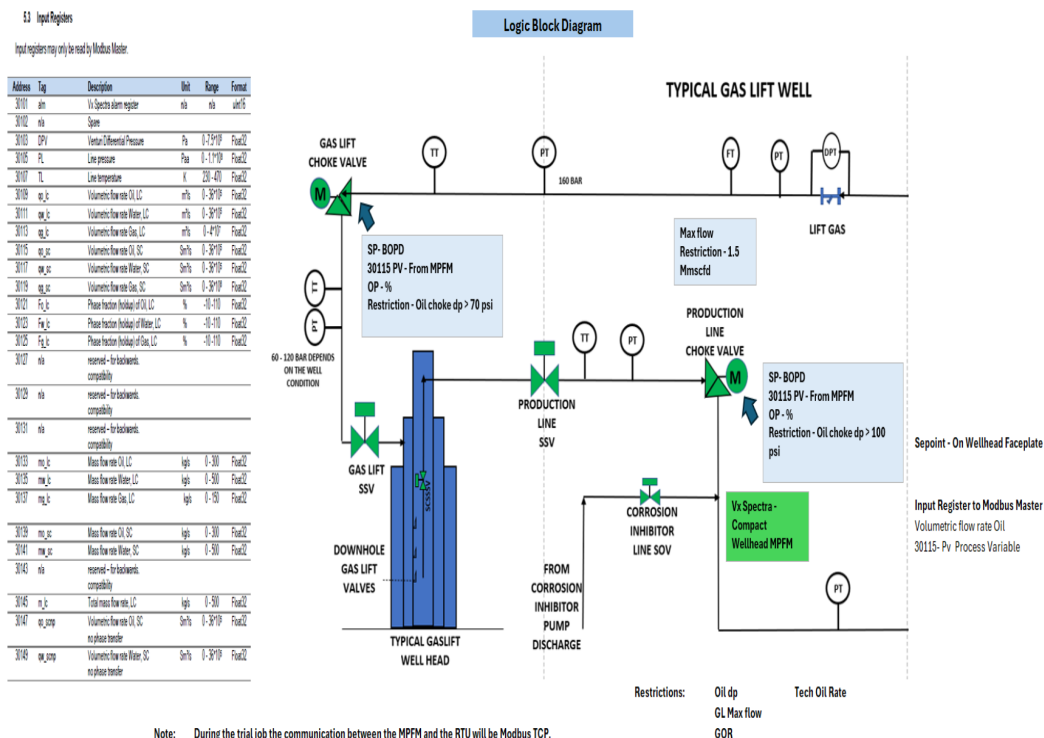


Figure 7—Typical Gas lift wellhead

MPFM Input Register

Input registers may only be read by Modbus Master.

Address	Tag	Description	Unit	Range	Format
30101	alm	Vx Spectra alarm register	n/a	n/a	uint16
30102	n/a	Spare			
30103	DPV	Venturi Differential Pressure	Pa	0 - 7.5×10^5	Float32
30105	PL	Line pressure	Paa	0 - 1.1×10^8	Float32
30107	TL	Line temperature	K	230 - 470	Float32
30109	qo_lc	Volumetric flow rate Oil, LC	m ³ /s	0 - 36×10^5	Float32
30111	qw_lc	Volumetric flow rate Water, LC	m ³ /s	0 - 36×10^5	Float32
30113	qg_lc	Volumetric flow rate Gas, LC	m ³ /s	0 - 4×10^7	Float32
30115	qo_sc	Volumetric flow rate Oil, SC	Sm ³ /s	0 - 36×10^5	Float32
30117	qw_sc	Volumetric flow rate Water, SC	Sm ³ /s	0 - 36×10^5	Float32
30119	qg_sc	Volumetric flow rate Gas, SC	Sm ³ /s	0 - 36×10^8	Float32
30121	Fo_lc	Phase fraction (holdup) of Oil, LC	%	-10 - 110	Float32
30123	Fw_lc	Phase fraction (holdup) of Water, LC	%	-10 - 110	Float32
30125	Fg_lc	Phase fraction (holdup) of Gas, LC	%	-10 - 110	Float32
30127	n/a	reserved – for backwards compatibility			
30129	n/a	reserved – for backwards compatibility			
30131	n/a	reserved – for backwards compatibility			
30133	mo_lc	Mass flow rate Oil, LC	kg/s	0 - 300	Float32
30135	mw_lc	Mass flow rate Water, LC	kg/s	0 - 500	Float32
30137	mg_lc	Mass flow rate Gas, LC	kg/s	0 - 150	Float32
30139	mo_sc	Mass flow rate Oil, SC	kg/s	0 - 300	Float32
30141	mw_sc	Mass flow rate Water, SC	kg/s	0 - 500	Float32
30143	n/a	reserved – for backwards compatibility			
30145	m_lc	Total mass flow rate, LC	kg/s	0 - 500	Float32
30147	qo_scnp	Volumetric flow rate Oil, SC no phase transfer	Sm ³ /s	0 - 36×10^5	Float32
30149	qw_scnp	Volumetric flow rate Water, SC no phase transfer	Sm ³ /s	0 - 36×10^5	Float32

Address	Tag	Description	Unit	Range	Format
30151	qg_scnp	Volumetric flow rate Gas, SC no phase transfer	Sm ³ /s	0 - 36*10 ⁸	Float32
30153	cst_sec	Start time cumulative values: Seconds	sec	0 - 59	uint16
30154	cst_min	Start time cumulative values: Minutes	min	0 - 59	uint16
30155	cst_hour	Start time cumulative values: Hours	hour	0 - 23	uint16
30156	cst_day	Start time cumulative values: Day	day	1 - 31	uint16
30157	cst_month	Start time cumulative values: Month	month	1 - 12	uint16
30158	cst_year	Start time cumulative values: Year	year	0 - 2100	uint16
30159	cvo_lc	Cumulated: Volumetric flow Oil, LC	m ³	0 - 3*10 ³⁸	Float32
30161	cvw_lc	Cumulated: Volumetric flow Water, LC	m ³	0 - 3*10 ³⁸	Float32
30163	cvg_lc	Cumulated: Volumetric flow Gas, LC	m ³	0 - 3*10 ³⁸	Float32
30165	cvo_sc	Cumulated: Volumetric flow Oil, SC	Sm ³	0 - 3*10 ³⁸	Float32
30167	cvw_sc	Cumulated: Volumetric flow Water, SC	Sm ³	0 - 3*10 ³⁸	Float32
30169	cvg_sc	Cumulated: Volumetric flow Gas, SC	Sm ³	0 - 3*10 ³⁸	Float32
30171	cmo_lc	Cumulated: Mass flow Oil, LC	kg	0 - 3*10 ³⁸	Float32
30173	cmw_lc	Cumulated: Mass flow Water, LC	kg	0 - 3*10 ³⁸	Float32
30175	cmg_lc	Cumulated: Mass flow Gas, LC	kg	0 - 3*10 ³⁸	Float32
30177	cmo_sc	Cumulated: Mass flow Oil, SC	kg	0 - 3*10 ³⁸	Float32
30179	cmw_sc	Cumulated: Mass flow Water, SC	kg	0 - 3*10 ³⁸	Float32
30181	n/a	reserved – for backwards. compatibility			
30183	cm_lc	Cumulated: Total mass flow, LC	kg	0 - 3*10 ³⁸	Float32
30185	cvo_scnp	Cumulated: Volumetric flow Oil, SC no phase transfer	Sm ³	0 - 3*10 ³⁸	Float32
30187	cvw_scnp	Cumulated: Volumetric flow Water, SC no phase transfer	Sm ³	0 - 3*10 ³⁸	Float32
30189	cvg_scnp	Cumulated: Volumetric flow Gas, SC no phase transfer	Sm ³	0 - 3*10 ³⁸	Float32
30191	WLR	Water Liquid Ratio (WLR), LC	%	-10 - 110	Float32
30193	GVF	Gas Volume Fraction (GVF), LC	%	-10 - 110	Float32
30195	GLR	Gas Liquid Ratio (GLR), LC	m ³ /m ³	0 - 36*10 ⁸	Float32
30197	BSW	Basic Sediment Water (BSW), SC	%	-10 - 110	Float32
30199	GOR	Gas Oil Ratio (GOR), SC	Sm ³ /S m ³	0 - 36*10 ⁸	Float32
30201	Do_lc	Density Oil, LC	kg/m ³	580 - 1100	Float32
30203	Dw_lc	Density Water, LC	kg/m ³	990 - 1400	Float32
30205	Dg_lc	Density Gas, LC	kg/m ³	4 - 700	Float32
30207	Do_sc	Density Oil, SC	kg/m ³	640 - 1100	Float32
30209	Dw_sc	Density Water, SC	kg/m ³	990 - 1400	Float32
30211	Dg_sc	Density Gas, SC	kg/m ³	0 - 2	Float32
30213	ts_sec	Vx Spectra time and date: Seconds	sec	0 - 59	uint16
30214	ts_min	Vx Spectra time and date: Minute	min	0 - 59	uint16

Address	Tag	Description	Unit	Range	Format
30215	ts_hour	Vx Spectra time and date: Hour	hour	0 - 23	uint16
30216	ts_day	Vx Spectra time and date: Day	day	1 - 31	uint16
30217	ts_month	Vx Spectra time and date: Month	month	1 - 12	uint16
30218	ts_year	Vx Spectra time and date: Year	year	0 - 2100	uint16
30219	w_prof	Vx Spectra well profile number	n/a	0 - 255	uint16
30220	N32	N 32 (LE) count rate – average	cps	0 - 100000	Float32
30222	N81	N 81 (HE) count rate – average	cps	0 - 100000	Float32
30224	N356	N 356 count rate – average	cps	0 - 100000	Float32
30226	NTotal	N Total count rate – average	cps	0 - 100000	Float32
30228	n/a	reserved – for backwards compatibility			
30230	n/a	reserved – for backwards compatibility			
30232	Mu_l_lc	Viscosity Liquid	Pa·s	0.0001 - 2	Float32
30234	Mu_o_lc	Viscosity Oil	Pa·s	0.0001 - 2	Float32
30236	OperatingPointLE	Operating Point LE	1/m	0 - 1000	Float32
30238	OperatingPointHE	Operating Point HE	1/m	0 - 1000	Float32
30240	OilPointLE	Triangle Oil Point LE	1/m	0 - 1000	Float32
30242	OilPointHE	Triangle Oil Point HE	1/m	0 - 1000	Float32
30244	WaterPointLE	Triangle Water Point LE	1/m	0 - 1000	Float32
30246	WaterPointHE	Triangle Water Point HE	1/m	0 - 1000	Float32
30248	GasPointLE	Triangle Gas Point LE	1/m	0 - 1000	Float32
30250	GasPointHE	Triangle Gas Point HE	1/m	0 - 1000	Float32
30252	Dm_lc	Density Mixture, LC	kg/m ³	4 - 1400	Float32
30254	DI_lc	Density Liquid, LC	kg/m ³	640 - 1400	Float32
30256	GOR1	Gas Oil Ratio 1, SC	Sm ³ /S m ³	0 - 3*10 ³⁸	Float32
30258	Dyn_DP	Dynamic DP	Pa	0 - 7.5*10 ⁵	Float32
30260	mhc_lc	Mass flow rate of hydrocarbons, LC	kg/s	0 - 500	Float32
30262	Pthroat	Pressure at Venturi throat	Paa	0 - 1.1*10 ⁸	Float32
30296	CGRAvg	Mass Condensate Gas Ratio Average	g/m ³	0 - 3*10 ³⁸	Float32

WH Choke & Gas lift Control Scenario

Maintaining The Oil Flow Rate

Pre-Requisite. Depending on current flow, first the well will be opened naturally – with validated oil rate well test. The same flow to be observed through the MPFM.

- dp must be more than 70 psi.(For GL mode)
- dp must be more than 100 psi.(For NF mode)

The setpoint will be given up to 1350 BOPD step by step with the increment of 50 BOPD and choke valve output to be observed.

Scenario-1

The following scenario to be used to achieve the above requirements.

1. Observe MPFM register address [30115 Volumetric flow rate Oil, SC] if the Oil flow rate less than [1350 bbl./d] for a period of [10 Mins.] then increase WH choke 10 % or 50 BOPD. The flow controller will keep on giving signal to oil choke opening until 100 psi differential across oil choke valve is achieved. (There must not be further signal to oil choke if output reaches 100% and dp is still more than 100 psi) - Condition to initiate signal to the oil choke valve (oil choke dp > 100 psi).
2. With the given setpoint 1350, the oil flow achieved less than SP, and if oil choke dp > 70psi (The selection tab to be switch to GL Mode). The command will go to gas lift choke valve to open gas flow until oil flow rate SP is accomplished 1350 BOPD – gas flow must not increase than 1.5 MMscfd (Restriction as per commissioning program) and gas choke valve downstream pressure must not increase more than 137 barg (as per WI requirement). And oil choke dp must not go less than 70 psi.

Scenario 2

- Operating mode – Gas lift
- New provision to be developed for operators to enter SP for volumetric flow rate
- MPFM register address [30115 Volumetric flow rate Oil, SC] to work as "PV"
- As per SP & PV gap, the output #1 to go to oil choke valve 10%. (Condition choke dp > 70 psi) – and observe at same output upto 5min.
- If SP & PV gas is still there, then same output increase command to be given to oil choke valve with interval of 10% and waiting for 5min at each interval.
- Once the 100 % output is achieved and oil choke valve is 100% open or oil choke dp is achieved 70 psi achieved (whichever first comes the interlock must work). There will no further output # 1 command to oil choke valve. Wait at this condition for 15min.
- If, the volumetric flow rate PV is still lesser than SP. Output # 1 will remain at this state, while output # 2 now to be given to gas choke valve to open.
- Gas flow condition – not more than 1.5 MMscfd
- Gas pressure condition – not more than 137 barg.
- Oil technical flow rate condition – must not increase.
- No more output command to gas choke valve if oil choke dp is achieved 70 psi or any of above conditions are met.

- Once Oil volumetric flow rate is achieved as per SP, no further command to gas lift choke valve.
- (Output # 2 – must be in accordance to PID/gain controller settings)
- Reverse action – if the oil volumetric flow rate PV is higher than SP
- Gas flow rate will be controlled/reduced as per output# 2 and PID controller settings.
- Once gas choke valve closes completely and still PV of volumetric flow rate is higher than SP, wait time delay for 15 min at the same condition...
- Then, command to go to oil choke valve from output # 1 to reduce at 10min interval / 10% reduction until SP is achieved.

General Benefits

By leveraging the capability to connect the MPFM with the piping connections available at the wellhead, the following benefits were achieved and will continue to be advantageous in the future:

- Optimization of manhours approximately 87 hours for 10 wells / 1,200 hours for 140 wells
- Production gain of 1,850,000 barrels of oil by preventing production loss from gas lift wells
- Optimization of GL flow rate, leading to savings of lean gas (9,000 MMscf for 50 wells) valued at 133 MMUSD
- Achievement of 100% well testing KPIs
- Effective and timely dual-string well testing resulting in a 10 MMUSD production gain
- Ensuring compliance with subsurface parameter restrictions, thereby preserving reservoir health
- Enhancement of well digitalization
- Facilitation of autonomous well control
- Real-time well testing
- Accurate representation of the well's behavior
- AI-based decision-making ensuring adherence to restrictions and WH/reservoir guidelines
- Reduction in GHG emissions due to less travel / commuting to well head
- Facilitation of autonomous well control with reservoir data– Next Phase (Integration with iPSO)

The implemented solution is highly effective in ensuring continuous production, timely well testing, and reducing manhours by leveraging autonomous control integrated with artificial intelligence. This methodology can be replicated across other existing facilities within the company and globally for the following reasons:

- Real-time flow monitoring and control
- Well testing compliance
- Effective reservoir management

Secured the real time monitoring of wells, required to ensure no alarm while approaching to well head as per recommendation in respective DGS.

Conclusion

The proposed size of the MPFM for the trial job (19 mm) has been selected based on the provided process data from the client for a potential well candidate. If any expectation of changing the well candidate, then another sizing study needs to be performed.

The proposed control scenarios can be updated based on the field specific needs.

During the trial job the communication between the MPFM and the RTU will be Modbus TCP.

Preserving critical functions & System effective availability of major equipment and during any operational or maintenance intervention by operator/maintenance technician, before proceeding to remote location, requisite resources can be proactively arranged which resulted in saving time and quality of intervention (Maintenance / Operations corrective actions) can be executed to avoid recurrences and achieve uninterrupted production. Successful concept deployment resulted in achieving multiple benefits for gas lift autonomous operation and can be deployed in ensuring effective & optimized wells operations.

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Glossary

MPP-	Multiphase pump
MPFM-	Multiphase fluid metering
BOPD-	Barrels Oil per day
BWPD-	Barrels water per day
WHCIP-	Well head closed in pressure
TSS-	Total suspended solids
PTS-	Potable Test Separator
MSM-	Multi Selector Manifold
KOD-	Knock out drum
FDIS-	Field Data Integration System
RCA-	Root Cause Analysis
DCS-	Distributed Control System
RDS-	Remote Degassing Station
CDS-	Central Degassing Station
F&G-	Fire & Gas
VMS-	Visual Management Board
PLC-	Programmable Logic Controller
KOD-	Knockout Drum
DGS-	Downgraded Situation
RTU-	Remote Telemetry Units
ACM-	Advanced Communication modules
ICS-	Instrumented control system
CHH-	Casing Head Housing
THS-	Tubing Head Spool

References

Gas lift wells Operating Manual, DCS, RTU and PI system data